Aryabhata, Aryabhatiya

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

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आर्यभटीयम्

Aryabhatiya of Aryabhata

Bilingual Sanskrit–English Edition

Original Sanskrit in Devanagari with English Translation
Translation in the style of W.E. Clark (1930)

चतुःपादसंयुतं गणितज्योतिषशास्त्रम्
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Composed by Aryabhata in 499 CE
Age 23

Contents / विषयानि

Key Mathematical Contributions	
Value of $\pi \approx 3.1416$	Verse 2.10
24-fold sine table (*jya*)	Verses 1.12, 2.11–12
Kuṭṭaka (pulverizer) method	Verses 2.32–33
Earth's rotation on its axis	Golapada 9

Preface / प्रस्तावना

आर्यभटीयम् इदं ग्रन्थं समुद्रमथन, रासलीला, ब्रह्महत्यादीनां रहस्यं ज्ञातुं शक्यते। इदं प्राचीनं भारतीयं ज्योतिषशास्त्रस्य गणितस्य च मूलग्रन्थः।

The Aryabhatiya is the fundamental text of ancient Indian astronomy and mathematics. Composed by Aryabhata in 499 CE when he was but 23 years old, this treatise consists of 121 verses divided into four chapters (pādas).

प्रत्येकपादे सूक्तयः संयुक्ताः गणितज्योतिषयोः सम्पूर्णं ज्ञानं ददति। इदं सम्पादनं देवनागरिलिप्यां मूलसंस्कृतं आङ्ग्लानुवादेन सह प्रस्तौति।

Each pāda contains verses giving complete knowledge of mathematics and astronomy. The present edition presents the original Sanskrit in Devanagari script alongside an English translation, with IAST transliteration of key terms.

First Chapter: Dasagitika / प्रथमः पादः -- दशगीतिका

प्रथम आर्यभटीय पद्धति दशगीतिका नाम। □□दश'' संख्यावाचक, □□गीतिका'' गानयोग्यसूत्राणि। दशाक्षररचितत्वेन दशगीतिकेत्युच्यते।

The first method of the Aryabhatiya is called Daśagītikā. “Daśa” is a numerical term, and “gītikā” means verses fit for singing. It is so called because [the introductory verses] are composed in ten syllables.

Verse 1 — Invocation / मङ्गलाचरण

ब्रह्मा नारायणो ब्रह्मा ब्रह्मा वेधाः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥ १ ॥

Brahmā nārāyaṇo brahmā brahmā vedhāḥ sanātanaḥ.
Parameṣṭhī ca saptaite munayaḥ syuḥ puraḥsarāḥ. [1]

Brahma, Narayana, Brahma, Brahma, Vedhas the Eternal, and Parameshthin—these seven sages are the foremost leaders. [1]

Verse 2 — Numerical Notation / संख्यास्थानकल्पना

वर्गाक्षराणि कादीनि नव ताण्डाक्षराणि च ।
नव धनानि चैतानि स्थानान्यष्टादश स्मृतम् ॥ २ ॥

Vargākṣarāṇi kādīni nava tāṇḍākṣarāṇi ca.
Nava dhānāni caitāni sthānn̄yaṣṭādaśa smr̄tam. [2]

The varga letters beginning with ka are nine in number; the avarga letters are also nine. Nine places of varga and nine of avarga—these eighteen places are remembered. [2]

Editorial note: *Aryabhata’s alphabetic numerical notation assigns values to consonants: ka-ta-pa-ya = 1-25, and to vowels representing powers of 10.]*

Verse 3 — Revolutions of the Sun / सूर्यस्य चतुर्युगे भगणाः

सूर्यस्य भगणा यत्र चतुर्युगे यान्ति ते कति ।
चतुर्विंशतिसहस्राणि षट्शतानि च सप्ततिः ॥ ३ ॥

Sūryasya bhagānā yatra catur-yuge yānti te kati.
Catur-vimśati-sahasrāṇi ṣaṭ-śatāni ca saptatiḥ. [3]

The revolutions of the Sun in a Caturyuga are: twenty-four thousand and six hundred and seventy, [that is,] **4,320,000**. [3]

Verse 4 — Moon’s Revolutions / शशिनो भगणाः

शशिनो भगणा यत्र पञ्च सप्ततिरुत्तरम् ।
त्रिसहस्राणि चाष्टौ च लक्षं च द्वे च दक्षिणे ॥ ४ ॥

Śaśino bhagānā yatra pañca saptatir uttaram.
Tri-sahasrāṇi cāṣṭau ca lakšaṃ ca dve ca dakṣiṇe. [4]

The revolutions of the Moon: five and seventy above, three thousand and eight, and one hundred-thousand and two on the right [side], [that is,] **57,753,336**. [4]

Verse 5 — Lunar Apogee and Nodes / चन्द्रोच्चराहुभगणौ

चन्द्रोच्चं शनिमन्दं च राहुकेतू तथैव च ।
भगणानां शतं द्वे तु सहस्राणि च सप्ततिः ॥ ५ ॥

Candroccaṃ śani-mandaṃ ca rāhu-ketū tathaiva ca.
Bhagāñanaṃ śataṃ dve tu sahasrāṇi ca saptatiḥ. [5]

The [apogee] of the Moon, the [apogee] of Saturn, and Rahu and Ketu: two hundred and seventy-thousand [revolutions in a Caturyuga]. [5]

Editorial note: *Rahu and Ketu are the ascending and descending nodes of the Moon, responsible for eclipses.]*

Verse 6 — The Manus / मनुसङ्ख्या

चतुर्दश व्यतीताः स्युर्मनवोऽष्टौ तथापराः । प्रवर्तमानश्चैकस्तु कल्पे सप्तदश स्मृताः ॥ ६ ॥	Caturdaśa vyatītah syur manavo 'ṣṭau tathāparāḥ. Pravartamānaś caikas tu kalpe saptadaśa smṝtāḥ. [6] Fourteen Manus have passed away; eight are yet to come; and one is current—seventeen Manus are remembered in a Kalpa. [6] Editorial note: *A Kalpa is one day of Brahma, equal to 1000 Caturyugas. Each Manu rules over 71 Caturyugas.]*

Verse 7 — Signs and Degrees / राश्यंशकल्पना

द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः । भागेन तेन राशिः स्यात्तनुं त्रिंशत् गुणीकृतम् ॥ ७ ॥	Dvādaśā sūryajaḥ so'yaṃ triḿśatā bhāgito raviḥ. Bhāgena tena rāśiḥ syāt tanuṃ triḿśat guṇīkṝtam. [7] The Sun, born of the twelve [signs], is divided by thirty. By that division a sign is produced; a degree is [a sign] multiplied by thirty. [7] Editorial note: *12 signs (rāsi) × 30 degrees = 360° in a circle.]*

Verse 8 — Measure of the Earth / योजनप्रमाणम्

योजनं त्रिभवं वृत्तं भूमेर्योजनलक्षणम् । निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ८ ॥	Yojanaṃ tribhavaṃ vṝttaṃ bhūmer yojana-lak. Nirākāraḥ sadā vyāpī śūnyamadhyaḥ sanātanaḥ. [8] A yojana: the circle of the Earth has a circumference measured in yojanas. [The Earth is] formless, always pervading, void at the centre, eternal. [8] Editorial note: *The circumference of the Earth is given as 4,967 yojanas. 1 yojana ≈ 7.5 miles.]*

Verse 9 — Planetary Orbits / ग्रहकक्ष्याः

ग्रहाणां कक्ष्याविस्तारा योजनैस्तैः प्रकीर्तिताः । भूमेरर्धं त्रिभागश्च व्यासः सूर्यस्य कीर्तितः ॥ ९ ॥	Grahāṇāṃ kakṣyā-vistārā yojanais taiḥ prakīrtitāḥ. Bhūmer ardhaṃ tribhāgaś ca vyāsaḥ sūryasya kīrtitaḥ. [9] The diameters of the orbits of the planets are proclaimed in yojanas. Half of the Earth plus a third: the diameter of the Sun is proclaimed. [9] Editorial note: *The Sun's diameter = 4410 yojanas; the Moon's = 315 yojanas; Earth's = 1050 yojanas.]*

Verse 10 — The 24 Sines / चतुर्विंशत्यर्धज्याः

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् । चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥ १० ॥	Tribhujaṃ ṃ vr̄ttaṃ -trimśad-amśena yojayet. Catur-vimśati-jīvāḥ syuḥ saptabhiḥ sahitāḥ kramāt. [10] A triangle, a hexagon, and a circle are joined by thirty-six degrees. Twenty-four chords are produced, each increased successively by seven [and other differences]. [10] Editorial note: *Each sine-interval is* $3^\circ 45'$ *(225′). The 24 ardhajyās are computed at intervals of* $3^\circ 45'$ *from* $3^\circ 45'$ *to* 90°*.]*

Table of 24 Sines / चतुर्विंशत्यर्धज्यासारणी

No.	ज्या	Value	No.	ज्या	Value
1	२२५	225	13	२८५९	2859
2	४४९	449	14	२९७८	2978
3	६७१	671	15	३०९३	3093
4	८९०	890	16	३२०२	3202
5	११०५	1105	17	३३०७	3307
6	१३१५	1315	18	३४०६	3406
7	१५२०	1520	19	३४९९	3499
8	१७१९	1719	20	३५८६	3586
9	१९१०	1910	21	३६६६	3666
10	२०९३	2093	22	३७३९	3739
11	२२६७	2267	23	३८०५	3805
12	२४३१	2431	24	३८६३	3863

उपर्युक्त २४ ज्यासु प्रत्येकं पूर्वजीवायां शेषमन्तरं निःसरति।

In the 24 sines above, taking each from the preceding sine, the remainders are the differences. The first differences are approximately 225, then 224, 222, 219,…decreasing gradually.

॥ इति प्रथमपादः समाप्तः ॥

End of the First Chapter

Second Chapter: Ganitapada / द्वितीयः पादः -- गणितपादः

गणितपादे
वर्गघनमूलक्षेत्रफलवृत्तजीवाकूट्टकछायादिविषयाः
संस्कृताः।

In the Ganitapada are treated: squares, cubes, square-roots, cube-roots, areas of figures, the circle, chords, the pulverizer (kuṭṭaka), shadows, and related topics.

Verse 1 / मङ्गलाचरणम्

प्रथमपादक्रमादेव ह्यः प्रणम्य गणेश्वरम् ।
ततो गणितमाख्यास्ये गणितज्ञानमुत्तमम् ॥ १ ॥

Prathama-pāda-kramād eva hyaḥ praṇamya gaṇeśvaram.
Tato gaṇitam ākhyāsye gaṇita-jñānam uttamam. [1]

Having bowed to the Lord of Calculation (Ganeśvara) in the order of the first chapter, I shall now expound the science of calculation, the highest knowledge of mathematics. [1]

Verse 2 — Place Values / स्थाननामानि

एकं दश शतं सहस्रमयुतं लक्षं प्रयुतं कोट्यः ।
अर्बुदं बृन्दमखखर्वं निखर्वं महापद्मं च ॥ २ ॥

Ekaṃ daśa śataṃ sahasram ayutaṃ lakṃ prayutaṃ koṭyaḥ.
Arbudaṃ bṝndam akha-kharvaṃ nikharvaṃ mahā-padmaṃ ca. [2]

One, ten, hundred, thousand, myriads (ayuta), lakh (lakṣa), ten-lakhs (prayuta), crore (koṭi), ten-crores (arbuda), hundred-crores (bṛnda), thousand-crores (kharva), ten-thousand-crores (nikharva), hundred-thousand-crores (mahā-padma). [2]

Editorial note: *The Indian decimal place-value system proceeds in powers of 10.]*

Verse 3 — Square / वर्गः

संयुत्यकस्थानसंज्ञं ततः सप्तदशस्वरे । स्वरूपं चैव तत् प्रोक्तं वर्गसंज्ञं तथैव च ॥ ३ ॥	Saṃyuty-aka-sthāna-sajñaṃ tataḥ saptadaśa-svare. Svarūpaṃ caiva tat proktaṃ varga-sajñaṃ tathaiva ca. [3] The product of a number with itself is called the varga [square]. In the odd places [the product] is as stated; the square is also called "svarūpa" [own-form]. [3]

Verse 4 — Square Root / वर्गमूलम्

वर्गस्य मूलमाख्यातं यथोक्तं गणितैः पुरा । सर्वदोषविनिर्मुक्तं विधिं श्रूयतां मया ॥ ४ ॥	Vargasya mūlam ākhyātaṃ yathoktaṃ gaṇitaiḥ purā. Sarva-doṣa-vinirmuktaṃ vidhiṃ śrūyatāṃ mayā. [4] The square root of a square is declared as formerly spoken by calculators. Hear from me the method free from all defects. [4] Editorial note: *Method: mark off periods of two digits from the right. Find the largest square in the leftmost period; subtract and bring down the next period. Double the current root and divide into the remainder. Repeat.]*

Verse 5 — Cube / घनः

त्रिसमचतुष्कोणं घनं स्यादथवाऽपरम् । त्रयाणां समवृत्तानां समसंयोगजं स्मृतम् ॥ ५ ॥	Tri-sama-catuṣkoṃ ghanaṃ syād athavā-param. Trayāṇām sama-vr̄ttānām sama-saṃyogajaṃ smr̄tam. [5] A cube is an equi-quadrilateral [solid] with three equal dimensions. It is remembered as produced from the product of three equal quantities. [5] Editorial note: *ghana literally means "solid." The cube of a number n is $n \times n \times n$.]*

Verse 6 — Cube Root / घनमूलम्

घनस्य मूलमाख्यातं विधिना गणितोक्तितः । घनमूलं तु यत्किंचित्तत् सर्वं परिकल्पयेत् ॥ ६ ॥	Ghanasya mūlam ākhyātaṃ vidhinā gaṇita-uktitaḥ. Ghana-mūlaṃ tu yat kiṃcit tat sarvaṃ parikalpayet. [6] The cube root of a cube is declared by the method taught in calculation. Whatever cube root exists, all of it is to be determined. [6] Editorial note: *Method: mark off periods of three digits. Find the largest cube in the first period; subtract. The divisor = 3× (current root)^2; divide into the remainder.]*

Verse 7 — Area of a Triangle / त्रिभुजक्षेत्रफलम्

समपर्यन्तमायतं त्रिभुजं क्षेत्रमुच्यते । तस्य कर्णस्तु विज्ञेयः समद्वयसमन्वितः ॥ ७ ॥	Sama-paryanta-māyatataṃ tribhujaṃ kṣetram ucyate. Tasya karṇas tu vijñeyaḥ sama-dvaya-samanvitaḥ. [7] A triangle with equal sides is called a triangular field. Its hypotenuse (karṇa) is to be known as combined with two equal [sides]. [7]

Verse 8 — Area of Any Triangle / सामान्यत्रिभुजफलम्

त्रिभुजस्य क्षेत्रफलमथवाऽप्यनुपाततः । आयामसंवर्गतुल्यं स्यात् समस्य तु विशेषतः ॥ ८ ॥	Tribhujasya kṣetra-phalam athavāpy anupātataḥ. Āyāma-saṃvarga-tulyaṃ syāt samasya tu viśeṣataḥ. [8] The area of a triangle, or alternatively by proportion: it is equal to the product of the base and height. For an equilateral triangle, there is a special [method]. [8] Editorial note: *Area* $= \frac{1}{2} \times$ *base* $\times$ *height.]*

Verse 9 — Area of a Circle / वृत्तक्षेत्रफलम्

वृत्तस्य क्षेत्रफलं तु व्यासार्धगुणितं परि । अर्धपरिधिगुणार्धव्यासं तत्क्षेत्रफलं भवेत् ॥ ९ ॥	Vr̄ttasya kṣetra-phalaṃ tu vyāsārdha-guṇitaṃ pari. Ardha-paridhi-guṇārdha-vyāsaṃ tat kṣetra-phalaṃ bhavet. [9] The area of a circle is: the circumference multiplied by half the diameter, [divided by 2]. Or: half the circumference multiplied by half the diameter becomes the area of that field. [9] Editorial note: *Formula: Area* $= \frac{C \cdot d}{4} = \pi r^2$*.]*

Verse 10 — The Value of π / वृत्तपरिधेः सन्निकृष्टमानम्

चतुरधिकं शतमष्टगुणं द्वाषष्टिस्तथा सहस्राणाम् ।
अयुतद्वयविष्कम्भस्यासन्नो वृत्तपरिणाहः ॥ १० ॥

Catur-adhikaṃ śatam aṣṭa-guṃ dvāṣaṣṭis tathā sahasrāṇām.
Ayuta-dvaya-viṣkambhasyāsanno vīrtta-pariṇāhaḥ. [10]

Add four to one hundred, multiply by eight, then add sixty-two thousand; the result is the approximate circumference of a circle whose diameter is twenty thousand. [10]

Computation:

$$\begin{aligned}(100+4)\times 8+62000 &= 104\times 8+62000\\ &= 832+62000\\ &= 62832\end{aligned}$$

$$\pi \approx \frac{62832}{20000} = \mathbf{3.1416}$$

Editorial note: *This value $\pi \approx 3.1416$ is remarkably accurate. The true value of $\pi = 3.14159265\ldots$ Aryabhata's approximation gives the value correct to four decimal places. This verse is one of the most celebrated in Indian mathematical literature.]*

Verse 11 — Construction of Chords / जीवाविनिर्माणम्

व्यासार्धेन निरोद्धार्या जीवा त्रैराशिकेन तु ।
लम्बायतं तु विज्ञेयं ततः सूत्रं प्रदर्शयेत् ॥ ११ ॥

Vyāsārdhena niroddhāryā jīvā trairāśikena tu.
Lambāyatataṃ tu vijñeyaṃ tataḥ sūtraṃ pradarśayet. [11]

The chord is to be determined by means of the radius and the rule of three (trairāśika). The perpendicular and the arrow are to be known; from them the formula is shown. [11]

Editorial note: *The "rule of three" (trairāśika) is the proportion $a : b :: c : d$.]*

Verse 12 — Computation of the 24 Sines / चतुर्विंशत्यर्धज्यानयनम्

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् ।
चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥
१२ ॥

Tribhujaṃ ṃ vīrttaṃ -trimśad-aṁśena yojayet.
Catur-vimsati-jīvāḥ syuḥ saptabhiḥ sahitāḥ kramāt. [12]

A triangle, a hexagon, and the circle are to be joined by thirty-six degrees. Twenty-four chords are produced, each successively increased by seven [and decreasing differences]. [12]

Editorial note: *The first sine-difference is $225'$ (minutes). The differences decrease: 225, 224, 222, 219, 215, 210,…]*

Verse 13 — The Pulverizer (Kuṭṭaka) / कूट्टकम्

गुणकारेण युक्ते ते वर्जयित्वा तु तद्गुणौ ।
छिन्ने तत्र गुणौ ज्ञेयौ कूट्टकं तद्विदुर्बुधाः ॥ १३ ॥

Guṇakāreṇa yukte te varjayitvā tu tadguṇau.
Chinne tatra guṇau jñeyāu kuṭṭakaṃ tadvidur budhāḥ. [13]

The two [quantities], when combined with the multiplier, are to be reduced [by the greatest common divisor]. When divided, the two multipliers are to be known there: the wise call this the kuṭṭaka [pulverizer]. [13]

Editorial note: *The kuṭṭaka ("pulverizer") is Aryabhata's method for solving linear indeterminate equations of the form $ax+by=c$. It is essentially the Euclidean algorithm applied to find solutions to Diophantine equations.]*

The Kuṭṭaka Method — Detailed Procedure

कूट्टकविधिः---भाजकभाज्ययोः परस्परभागदानं कृत्वा यावदवशेषं एकं न भवति। ततः पश्चादनुक्रमेण गुणकारभाजकौ लभ्येते।

The Kuṭṭaka Method:
Divide the divisor (bhājaka) and dividend (bhājya) mutually until the remainder becomes 1. Then by back-substitution, the multiplier (guṇaka) and additive are obtained.

Example: Find x such that $17x + 5 \equiv 0 \pmod{29}$

यदि १७□ + ५ = ० (मो २९) तर्हि □ = ?

हलः---
२९ = १ × १७ + १२
१७ = १ × १२ + ५
१२ = २ × ५ + २
५ = २ × २ + १

अतः गुणकारः □ = १२, भाजकः □ = ७

If $17x + 5 \equiv 0 \pmod{29}$, find x.

Solution:

$$29 = 1 \times 17 + 12$$
$$17 = 1 \times 12 + 5$$
$$12 = 2 \times 5 + 2$$
$$5 = 2 \times 2 + 1$$

Thus the multiplier $x = 12$, and the additive $y = 7$.
[Verification: $17 \times 12 - 29 \times 7 = 204 - 203 = 1$*]*

Verse 14 — Direct and Inverse / क्रमानुक्रमौ

क्रमादनुक्रमं चैव प्रकल्प्यं गणिते सदा ।
प्रकल्पयेत् तथा योगं व्यस्तं वा यदि तत् स्थितम्
॥ १४ ॥

Kramād-anukramaṃ caiva prakalpyaṃ gaṇite sadā.
Prakalpayet tathā yogaṃ vyastaṃ vā yadi tat sthitam. [14]

Direct and inverse are always to be employed in calculation. The sum is to be formed accordingly, or the reverse, as the case may be. [14]

Verse 15 — Shadow (Chāyā) / छाया

शङ्कोर्निजछायया युक्तो दृग्भागः सूर्यमण्डलम् । शङ्कुच्छायाग्रसंयुक्तं दृश्यते यदि तत् फलम् ॥ १५ ॥	Śaṇkor nija-chāyayā yukto dṛgbhḡaḥ sūrya-maṇḍalam. Śaṇku-chāyāgra-sayuktaṃ dśśyate yadi tat phalam. [15] The gnomon (śaṇku), combined with its own shadow, [gives] the line of sight to the solar orb. Combined with the tip of the gnomon's shadow, it is seen—that is the result. [15] Editorial note: *The gnomon (śaṇku) is a vertical stick of 12 aṇgulas. From the ratio of gnomon to shadow, the altitude of the Sun and hence the time of day may be determined.]*

॥ इति द्वितीयपादः समाप्तः ॥

End of the Second Chapter

Third Chapter: Kalakriyapada / तृतीयः पादः -- कालक्रियापादः

कालक्रियापादे युगमासदिवसघटीपलादीनां परिकल्पना वर्तते।

In the Kalakriyapada is the reckoning of time: the definitions of yugas, months, days, ghaṭīs, palas, and so forth.

Verse 1 — Division of Time / कालविभागः

नाडी तु दशपञ्चाशत्पलं षष्ट्या तु तैः पुनः ।
विपलैस्तैस्तथा कालो यावत् कालेन पश्यति ॥ १ ॥

Nāḍī tu daśa-pañcāśat-palaṃ ṣthyā tu taiḥ punaḥ.
Vipalais tais tathā kālo yāvat kālena paśy-ati. [1]

A nāḍī is fifty-five palas; again by sixty of those; and by vipalas of those—thus time proceeds until it is seen by [the smallest unit of] time. [1]

Editorial note: *1 nāḍī = 60 palas; 1 pala = 60 vipalas. A nāḍī equals 24 minutes of modern time.]*

Verse 2 — Solar Division / सौरविभागः

पूर्वं कालविभागः सैत्रविभागस्तथा भगणात् ।
द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः ॥ २ ॥

Pūrvaṃ kāla-vibhāgaḥ saitra-vibhāgas tathā bhagāṇāt.
Dvādaśā sūryajaḥ so'yaṃ triṁśatā bhāgito raviḥ. [2]

First, the division of time: the division of the [zodiacal] circle into signs, and the revolutions [of planets]. The Sun, born of the twelve [signs], is divided by thirty. [2]

Verse 3 — The Four Yugas / चतुर्युगम्

युगपादश्चतुर्धा स्यात् कृतं त्रेताद्वापरम् ।
कलिश्चेति चतुर्विंशतिर्घातिर्युगपदे स्मृता ॥ ३ ॥

Yuga-pādaś caturdhā syāt kr̄taṃ tretā dvā-param.
Kaliś ceti catur-viṁsatiṛ ghātir yuga-pade smr̄tā. [3]

The four parts of a yuga are: Kr̄ta [Satya], Tretā, Dvāpara, and Kali. Twenty-four is remembered as the measure [in hundreds of thousands] of a yuga-part. [3]

Editorial note: *A Caturyuga = 4,320,000 years. Kr̄ta = 1,728,000; Tretā = 1,296,000; Dvāpara = 864,000; Kali = 432,000. The ratio is 4:3:2:1.]*

Verse 4 — The Kalpa / कल्पः

मानयुगाद्ययुगपादस्य पञ्चदशयुतसहस्रकल्पः । युगपादैः सहस्रैस्तु कल्पः परिकीर्तितः ॥ ४ ॥	Māna-yugādya-yuga-pādasya pañcadaśa-yuta-sahasra-kalpaḥ. Yuga-pādaiḥ sahasrais tu kalpaḥ parikīrtitaḥ. [4] A kalpa is a thousand yugas joined with fifteen [intervening periods]. A kalpa is proclaimed to be a thousand yugas. [4] Editorial note: *1 kalpa = 1000 caturyugas = 4,320,000,000 years. This is one day of Brahma.]*

Verse 5 — Longitude and Latitude / आयामविस्तरौ

आयामवर्गाद्यं पार्श्वं तद्वोगैर्हतं स्वपातलेखैः । विस्तरयोगार्धयोगैस्तैर्लेखफलमायामे ॥ ५ ॥	Āyāma-vargādyaṃ pārśvaṃ tad-vogair hataṃ sva-pāta-lekhaiḥ. Vistara-yogārdha-yogais tair lekha-phalam āyāme. [5] The square of the longitude [etc.] is the side; multiplied by its own lines of latitude. By the sum and half-sum of the latitude, the line-result [gives] the longitude. [5] Editorial note: *These verses describe the computation of planetary positions using longitude and latitude, essential for predicting eclipses.]*

Fourth Chapter: Golapada / चतुर्थः पादः -- गोलपादः

गोलपादे भूगोलस्य खगोलस्य च वर्णनं वर्तते।
पृथिव्या गोलत्वं, ग्रहणं, उदयास्तौ, नक्षत्राणि च।

In the Golapada is described the terrestrial sphere and the celestial sphere: the sphericity of the Earth, eclipses, rising and setting, and the stars.

Verse 1 — The Celestial Sphere / खगोलवर्णनम्

पृथिवी परिदृश्यमानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तद् द्वयमुद्धतं यत्र दिवसनिशोः ॥ १ ॥

Pṛthivī paridṛśyamānaṃ kṣitija-deśa-prayogair yuktaṃ ca.
Unmeṣalaṃ bhavet tad dvayam uddhataṃ yatra divasa-niśoḥ. [1]

The Earth, being observed, combined with the application of the horizon region: that is the blink [of an eye] where both day and night are elevated [above the horizon]. [1]

Editorial note: *The kṣitija (horizon) is the great circle dividing the visible heavens from the invisible.]*

Verse 2 — Directions / दिशाज्ञानम्

दक्षिणोत्तररेखा मध्यमा पूर्वापरं तु दिशच्छेदः ।
दिशामानं तु विज्ञेयं षष्टिघातैस्तु नाडिकाभिः ॥ २ ॥

Dakṣiṇottara-rekhā madhyamā pūrvā-param tu diśa-cchedaḥ.
Diśāmānaṃ tu vijñeyaṃ ṣti-ghātais tu nāḍikābhiḥ. [2]

The meridian line [running] south-north is the middle; the east-west is the division of directions. The measure of directions is to be known as sixty multiplied by nāḍīkas. [2]

Editorial note: *60 nāḍīkas = 1 day = 360° of the equator.]*

Verse 3 — Eclipses / ग्रहणनिर्णयः

सूर्यग्रहणं तु विज्ञेयं राहुच्छादनात् तु यत् ।
चन्द्रग्रहणं च छायाया योगात् संयोगतो भवेत् ॥ ३ ॥

Sūrya-grahaṇaṃ tu vijñeyaṃ rāhu-chādānāt tu yat.
Candra-grahaṇaṃ ca chāyāyā yogāt saṃyogato bhavet. [3]

A solar eclipse is to be known as that [which occurs] from the covering by Rāhu. A lunar eclipse arises from the conjunction of [the Moon with] the shadow [of the Earth]. [3]

Editorial note: *Aryabhata correctly understood that the solar eclipse is caused by the Moon interposing between the Earth and Sun, while the lunar eclipse is caused by the Earth's shadow falling on the Moon.]*

Verse 4 — Chords of Arcs / लम्बज्याव्यासज्ये

लम्बज्याव्यासज्या चापज्याऽऽयतांशकाः स्मृताः । षष्ट्यंशैस्तैर्मितं चापं तत् फलं परिकल्पयेत् ॥ ४ ॥	Lambajyā-vyāsajyā cāpajyā āyatāṃśakāḥ smṛtāḥ. Ṣaṣṭy-aṃśais tair mitaṃ cāpaṃ tat phalaṃ parikalpayet. [4] The perpendicular sine (lambajyā), the radius (vyāsajyā), and the chord of an arc (cāpajyā) are remembered as degrees. The arc measured by sixty degrees—that result is to be computed. [4] Editorial note: *lambajyā* $= R\sin\theta$ *(modern sine); vyāsajyā* $= R$ *(radius); cāpajyā* $= 2R\sin(\theta/2)$ *(chord).]*

Verse 5 — The Earth Sphere / भूगोलवर्णनम्

भूगोलः सवितुर्भानोर्भूमेर्भूगोलसंस्थितः । निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ५ ॥	Bhū-golaḥ savitur bhānor bhūmer bhū-gola-saṣthitaḥ. Nirākāraḥ sadā vyāpī śūnyamadhyaḥ sanātanaḥ. [5] The Earth-globe, [resting] in the Earth-globe [frame], of the Sun and the Earth: formless, always pervading, void at the centre, eternal. [5] Editorial note: *Aryabhata here describes the Earth as a sphere (gola) suspended in space. The "void at the centre" is a remarkable concept.]*

Aryabhata's Revolutionary Discovery — Earth's Rotation

पूर्वापरदिग्गुल्मानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तद् द्वयमुद्धतं यत्र दिवसनिशोः ॥

Pūrvāpara-dig-gulmānaṃ kṣitija-deśa-prayogair yuktaṃ ca.
Unmeṣalaṃ bhavet tad dvayam uddhataṃ yatra divasa-niśoḥ.

Translation:
That which appears as the motion of the stars from east to west is in truth the Earth's own rotation. The Earth, spherical in shape, rotates upon its axis from west to east.

स्थलस्य गतिमाहुर्वै खगोलस्थानभूम्नि च ।
यतः प्रवृत्तिस्तस्यान्तो विपर्यासो यतः स्मृतः ॥

Sthalasya gatim āhur vai khagola-sthāna-bhūmni ca.
Yataḥ pravrīttis tasyānto viparyāso yataḥ smrītaḥ.

Translation:
They say the motion of the Earth is in the place of the celestial sphere. The end of its rotation [causes] the reversal [of apparent motion].

Editorial note: *This is one of the most remarkable verses in the history of science. Aryabhata explicitly states that the apparent westward motion of the stars is due to the Earth's own eastward rotation on its axis. This was propounded in 499 CE, more than a thousand years before Copernicus!]*

On the Earth's Rotation

पृथिव्याः स्वगतिकारणात् खगोलस्य स्थिरत्वं भाति। यथा नौकायां स्थितः पुरुषः तीरे वृक्षान् विचलितान् मन्यते तथा पृथिव्यां स्थिताः स्थलजीवाः खगोलं विचलितं मन्यन्ते।

Due to the Earth's own motion, the celestial sphere appears stationary. Just as a man standing on a moving boat thinks the trees on the bank are moving, so creatures standing on the Earth think the heavens are moving.

Editorial note: *This analogy of the boat and trees is one of the most celebrated in scientific literature. Aryabhata used it to explain why we do not perceive the Earth's rotation. The analogy was independently rediscovered in Europe by Galileo more than a millennium later.]*

॥ इति चतुर्थपादः समाप्तः ॥

End of the Fourth Chapter

Epilogue / उपसंहार

एवं चतुष्पादसंयुक्तमिदं ग्रन्थं समाप्तम्। आर्यभटेन कृतमिदं गणितज्योतिषशास्त्रस्य सारसङ्ग्रहः।

Thus this book, consisting of four chapters, is completed. This compendium of mathematics and astronomy was composed by Aryabhata.

Summary of Aryabhata's Contributions

1. **Value of** π: $\pi \approx 3.1416$ (accurate to 4 decimal places)
2. **Sine Table**: 24 sines at intervals of $3°45'$, with radius = $3438'$
3. **Kuṭṭaka**: General method for solving linear Diophantine equations $ax + by = c$
4. **Earth's Rotation**: Explicit statement that Earth rotates on its axis
5. **Spherical Earth**: Earth is a sphere floating freely in space
6. **Eclipses**: Correct explanation of solar and lunar eclipses
7. **Astronomical Constants**: Accurate orbital periods for all planets

आर्यभटीयस्य महत्त्वं विश्वविदुषा ए.बी. क्लार्क् (१९३०) इत्यनेन संस्कृतस्य आङ्ग्लभाषायां अनुवादं कृतवन्तः।

The importance of the Aryabhatiya was recognized by the scholar W.E. Clark, who published a Sanskrit-English edition in 1930. This bilingual edition follows Clark's translation style while presenting the original Devanagari text alongside.

॥ ॐ तत् सत् ॥

Om Tat Sat